WEB DEVELOPMENT ACTIVITY-BASED LEARNING

Three case studies with application, theory, features, expected output, and source code

Case Study 1
Personal Portfolio Website

Case Study 2
College Information & Admissions Website

Case Study 3
Online Learning Platform Interface

Prepared as a learning reference. Each example is a self-contained HTML file with embedded CSS and JavaScript. Save the code as index.html and open it in a modern browser.

How to Use This PDF

Choose any three case studies as required by the activity. The included examples use HTML for page structure, CSS for layout and animation, and a small amount of JavaScript for interaction. The content is sample content and can be edited for your own institution, profile, or course catalogue.

Steps to run the code

1. Open a text editor such as Notepad or Visual Studio Code.
2. Copy one complete case-study code block into a new file.
3. Save the file as **index.html** (choose a separate folder for each project).
4. Double-click the file to open it in a browser.
5. Resize the browser window to test the responsive layout.

Note: These are front-end demonstrations. Forms and buttons show client-side interactions only; real admissions, authentication, payments, and course enrolment require a backend and appropriate data protection.

1. Personal Portfolio Website

Application

A portfolio website presents a student's or developer's background in one place. It can be shared with internship coordinators, recruiters, faculty members, and project collaborators. Replace the sample name, qualifications, skills, projects, and contact details with accurate information.

Theory

The page uses semantic HTML sections, a responsive CSS grid, smooth scrolling, hover transitions, and a short entrance animation. A mobile-friendly navigation layout helps the same page work on phones, tablets, and desktops. The contact link uses an email URL; no information is submitted to a server.

Features

Profile/hero section with a call-to-action • Education, technical skills, hobbies, projects, and achievements • Resume link placeholder and contact link • Responsive cards, hover effects, and entrance animation • Navigation links scroll to page sections

Expected Output

A single-page portfolio with a dark blue hero banner, profile introduction, section cards, and a contact area. On narrow screens, cards stack vertically and the navigation remains readable.

Source Code (HTML + CSS + JavaScript)

Copy the complete code below into a file named **index.html**.

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Personal Portfolio</title>
<style>
:root{--ink:#17324d;--blue:#176b87;--pale:#eef6fa;--white:#fff}
*{box-sizing:border-box} html{scroll-behavior:smooth}
body{margin:0;font:16px/1.6 Arial,sans-serif;color:var(--ink);background:#f7fafc}
header{background:var(--ink);color:white;padding:14px
  6%;display:flex;justify-content:space-between;align-items:center;gap:16px;flex-wrap:wrap}
nav{display:flex;gap:14px;flex-wrap:wrap}nav a{color:white;text-decoration:none}
nav a:hover{color:#8ee3ef}
.hero{text-align:center;padding:60px
  20px;background:linear-gradient(135deg,#17324d,#176b87);color:white;animation:appear .8s ease}
.hero h1{font-size:clamp(2rem,5vw,3.4rem);margin:0}
.button{display:inline-block;background:#8ee3ef;color:#17324d;padding:10px
  18px;border-radius:24px;text-decoration:none;font-weight:bold}
main{max-width:1050px;margin:auto;padding:24px}
section{scroll-margin-top:20px;margin:22px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:16px}
.card{background:white;padding:18px;border-radius:12px;box-shadow:0 4px 14px
  #17324d12;transition:transform .2s,box-shadow .2s}
.card:hover{transform:translateY(-5px);box-shadow:0 8px 22px #17324d22}
h2{color:var(--blue)}footer{text-align:center;background:var(--ink);color:white;padding:18px}
@keyframes appear{from{opacity:0;transform:translateY(12px)}to{opacity:1;transform:translateY(0)}}
@media(max-width:600px){header{align-items:flex-start}nav{gap:9px}.hero{padding:42px
  16px}main{padding:16px}}
</style>
</head>
<body>
<header><strong>Alex Morgan | Portfolio</strong>
<nav aria-label="Main navigation">
<a href="#about">About</a><a href="#education">Education</a><a href="#skills">Skills</a>
<a href="#projects">Projects</a><a href="#contact">Contact</a>
</nav></header>
<div class="hero" id="about">
<h1>Hello, I'm Alex Morgan</h1>
<p>Computer Science student | Front-end developer in training</p>
<a class="button" href="#projects">Explore my work</a>
</div>
<main>
<section id="education"><h2>Education</h2><div class="card">
<b>B.Sc. / B.Tech. in Computer Science</b><p>Example University · 2023–2027</p>
<p>Relevant topics: web development, databases, and programming.</p></div></section>
```

```
<section id="skills"><h2>Technical Skills</h2><div class="grid">
<div class="card"><h3>Web</h3><p>HTML, CSS, responsive design</p></div>
<div class="card"><h3>Programming</h3><p>JavaScript, Python, Java (beginner)</p></div>
<div class="card"><h3>Tools</h3><p>Git, VS Code, browser developer tools</p></div>
</div></section>
<section><h2>Hobbies &amp;amp; Achievements</h2><div class="grid">
<div class="card"><h3>Hobbies</h3><p>Reading, sketching, and solving coding problems.</p></div>
<div class="card"><h3>Achievements</h3><p>Completed a web design workshop and a student project
  showcase.</p></div>
</div></section>
<section id="projects"><h2>Projects</h2><div class="grid">
<div class="card"><h3>Student Task Planner</h3><p>A simple interface to organise assignments and
  deadlines.</p></div>
<div class="card"><h3>Campus Guide</h3><p>A responsive page listing useful campus locations.</p></div>
</div></section>
<section><h2>Resume</h2><div class="card"><p>Add your PDF resume to this project folder and update
  the filename below.</p>
<a class="button" href="resume.pdf" target="_blank" rel="noopener">View resume</a></div></section>
<section id="contact"><h2>Contact</h2><div class="card">
<p>Email: alex@example.com</p><a class="button" href="mailto:alex@example.com">Send an
  email</a></div></section>
</main>
<footer>© 2026 Alex Morgan · Personal Portfolio</footer>
</body></html>
```

2. College Information and Admissions Website

Application

A college website helps prospective students and families learn about the institution, compare departments, understand the admissions process, explore campus facilities, and find contact information. The sample content should be replaced with verified details before publication.

Theory

The page organises information into reusable cards and clear sections. A flexible navigation bar, hero banner, responsive grid, and CSS hover states improve readability. JavaScript provides a simple admissions notice interaction. A production site should connect the application form to a secure backend and publish current, accessible information.

Features

Institution name, hero banner, and navigation • About section and department/course cards • Admissions steps and campus facilities • Events and contact information • Responsive grid and interactive admissions notice

Expected Output

A college landing page with a prominent welcome banner, course cards, admissions steps, facilities, events, and contact details. Cards automatically change from multiple columns to one column on small screens.

Source Code (HTML + CSS + JavaScript)

Copy the complete code below into a file named **index.html**.

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1">
<title>Greenfield College</title>
<style>
*{box-sizing:border-box}body{margin:0;font:16px/1.6 Arial,sans-serif;color:#203040;background:#f7f9fb}
header{background:#123c35;color:white;padding:14px
  5%;display:flex;justify-content:space-between;align-items:center;flex-wrap:wrap;gap:12px}
nav{display:flex;gap:16px;flex-wrap:wrap}nav a{color:white;text-decoration:none}nav
  a:hover{text-decoration:underline;color:#b9f3d0}
.hero{padding:64px 7%;color:white;background:linear-gradient(120deg,#123c35,#21866b)}
.hero h1{font-size:clamp(2rem,5vw,3.2rem);margin:0}
.wrap{max-width:1100px;margin:auto;padding:24px}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:16px}
.card{background:white;padding:18px;border-radius:10px;border-top:4px solid #21866b;box-shadow:0 3px
  12px #102a4312}
h2{color:#17634f}.cta{border:0;border-radius:6px;padding:11px
  16px;background:#f2c14e;color:#24313b;font-weight:bold;cursor:pointer}
.cta:hover{filter:brightness(.95)}footer{background:#123c35;color:white;text-align:center;padding:20px}
@media(max-width:600px){.hero{padding:42px 20px}.wrap{padding:16px}}
</style></head>
<body>
<header><strong>Greenfield College</strong><nav aria-label="Main navigation">
<a href="#about">About</a><a href="#courses">Courses</a><a href="#admissions">Admissions</a>
<a href="#campus">Campus</a><a href="#contact">Contact</a></nav></header>
<div class="hero"><h1>Learn Today. Lead Tomorrow.</h1>
<p>Discover your interests, build practical skills, and prepare for the future.</p>
<button class="cta" onclick="showAdmissions()">Admissions information</button></div>
<main class="wrap">
<section id="about"><h2>About the College</h2><p>Greenfield College is a sample institution focused
  on academic learning, practical projects, and student development. Replace this paragraph with the
  college's approved history, mission, accreditation, and verified facts.</p></section>
<section id="courses"><h2>Departments &amp; Courses</h2><div class="grid">
<article class="card"><h3>Computer Science</h3><p>Programming, databases, web technologies, and
  software engineering.</p></article>
<article class="card"><h3>Business Studies</h3><p>Management, accounting, entrepreneurship, and
  communication.</p></article>
<article class="card"><h3>Life Sciences</h3><p>Biology, laboratory methods, and environmental
  studies.</p></article>
</div></section>
<section id="admissions"><h2>Admission Process</h2><div class="grid">
<article class="card"><h3>1. Explore</h3><p>Review eligibility criteria and available
  programmes.</p></article>
<article class="card"><h3>2. Apply</h3><p>Complete the official application and provide required
  documents.</p></article>
```

```
<article class="card"><h3>3. Confirm</h3><p>Follow official instructions for selection and
  enrolment.</p></article>
</div><p><b>Important:</b> Check the official college office for deadlines, fees, and
  eligibility.</p></section>
<section id="campus"><h2>Campus Facilities</h2><div class="grid">
<div class="card"><h3>Library</h3><p>Study spaces and learning resources.</p></div>
<div class="card"><h3>Laboratories</h3><p>Practical learning and project work.</p></div>
<div class="card"><h3>Sports &amp;amp; Clubs</h3><p>Activities that support student wellbeing and
  teamwork.</p></div>
</div></section>
<section><h2>Upcoming Events</h2><div class="grid">
<div class="card"><h3>Orientation Day</h3><p>Meet faculty and explore campus. Date: To be
  announced.</p></div>
<div class="card"><h3>Project Expo</h3><p>Student demonstrations and presentations. Date: To be
  announced.</p></div>
</div></section>
<section id="contact"><h2>Contact</h2><div class="card">
<p>Email: admissions@example.edu | Phone: +91-00000-00000</p>
<p>Address: Replace with the verified campus address.</p></div></section>
</main>
<footer>© 2026 Greenfield College · Sample educational website</footer>
<script>
function showAdmissions(){
  alert("Please check the official college notice for current eligibility, deadlines, and
    application links.");
}
</script></body></html>
```

3. Online Learning Platform Interface

Application

An online learning interface lets learners discover courses, review instructors and learning objectives, check progress, and begin a course. This example is suitable as a front-end prototype for a learning management system or course catalogue.

Theory

The design uses course cards, progress elements, responsive layout, and a JavaScript filter. Semantic buttons and accessible labels improve usability. The sample progress bars are illustrative; real progress should be stored for each learner by an authenticated backend.

Features

Course categories and course cards • Instructor names, descriptions, and learning objectives • Enrollment buttons with a demo confirmation • Progress indicators • Category filter and responsive layout

Expected Output

A learning dashboard showing course cards with category labels, instructor information, learning outcomes, progress bars, and enrolment buttons. Selecting a category filters the displayed courses; the enrolment button shows a demo message.

Source Code (HTML + CSS + JavaScript)

Copy the complete code below into a file named **index.html**.

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8"><meta name="viewport" content="width=device-width, initial-scale=1">
<title>LearnSpace</title>
<style>
*{box-sizing:border-box}body{margin:0;font:16px/1.55 Arial,sans-serif;background:#f4f7fb;color:#202d40}
header{padding:16px 5%;background:#252b5c;color:white;display:flex;justify-content:space-between;ali
  gn-items:center;gap:12px;flex-wrap:wrap}
header a{color:white;text-decoration:none}.hero{padding:38px 5%;background:#e5e9ff}
.hero h1{margin:0;color:#252b5c;font-size:clamp(2rem,4vw,3rem)}
.wrap{max-width:1150px;margin:auto;padding:24px}
.filters{display:flex;gap:10px;flex-wrap:wrap;margin:12px 0 20px}
button{cursor:pointer;border:0;border-radius:7px;padding:10px
  14px;background:#4e5bd5;color:white;font-weight:bold}
button:hover{background:#303b9b}
.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(250px,1fr));gap:18px}
.card{background:white;border-radius:12px;overflow:hidden;box-shadow:0 4px 16px #1c2b4a14}
.thumb{padding:28px 18px;background:linear-gradient(135deg,#d8f3ed,#dfe4ff);font-size:1.4rem;font-we
  ight:bold;color:#252b5c}
.content{padding:18px}.tag{font-size:.8rem;color:#4e5bd5;font-weight:bold}
progress{width:100%;height:12px;accent-color:#4e5bd5}
.note{min-height:24px;color:#17634f;font-weight:bold}
footer{text-align:center;padding:18px;background:#252b5c;color:white}
@media(max-width:600px){.wrap{padding:16px}.hero{padding:28px 18px}}
</style></head>
<body>
<header><strong>LearnSpace</strong><a href="#courses">Browse courses</a><span>Learning
  dashboard</span></header>
<div class="hero"><h1>Build a skill, one lesson at a time.</h1>
<p>Explore practical courses and keep track of your learning progress.</p></div>
<main class="wrap" id="courses">
<h2>Explore Courses</h2>
<div class="filters" aria-label="Filter courses">
<button onclick="filterCourses('all')">All</button>
<button onclick="filterCourses('Web')">Web Development</button>
<button onclick="filterCourses('Design')">Design</button>
<button onclick="filterCourses('Data')">Data</button>
</div>
<p id="message" class="note" role="status"></p>
<div class="grid">
<article class="card" data-category="Web"><div class="thumb">HTML &amp;amp; CSS Basics</div><div
  class="content">
<span class="tag">WEB DEVELOPMENT</span><h3>Build Your First Webpage</h3>
<p><b>Instructor:</b> Priya Sharma</p><p>Learn page structure, styling, and responsive layouts.</p>
<p><b>Learning objectives:</b> Create semantic HTML and style a flexible page.</p>
```

```
<label for="p1">Course progress: 35%</label><progress id="p1" value="35" max="100">35%</progress>
<p><button onclick="enroll('Build Your First Webpage')">Enroll / Continue</button></p></div></article>
<article class="card" data-category="Design"><div class="thumb">UI Design</div><div class="content">
<span class="tag">DESIGN</span><h3>Design Better Interfaces</h3>
<p><b>Instructor:</b> Omar Khan</p><p>Explore spacing, contrast, typography, and usability.</p>
<p><b>Learning objectives:</b> Apply visual hierarchy and build a simple wireframe.</p>
<label for="p2">Course progress: 60%</label><progress id="p2" value="60" max="100">60%</progress>
<p><button onclick="enroll('Design Better Interfaces')">Enroll / Continue</button></p></div></article>
<article class="card" data-category="Data"><div class="thumb">Data Essentials</div><div class="content">
<span class="tag">DATA</span><h3>Introduction to Data</h3>
<p><b>Instructor:</b> Maya Rao</p><p>Understand tables, basic analysis, and data visualisation.</p>
<p><b>Learning objectives:</b> Read a dataset and explain simple patterns.</p>
<label for="p3">Course progress: 10%</label><progress id="p3" value="10" max="100">10%</progress>
<p><button onclick="enroll('Introduction to Data')">Enroll / Continue</button></p></div></article>
</div></main>
<footer>© 2026 LearnSpace · Front-end demonstration</footer>
<script>
function filterCourses(category){
  document.querySelectorAll(".card[data-category]").forEach(card=>{
    card.hidden = category !== "all" && card.dataset.category !== category;
  });
  document.getElementById("message").textContent =
    category === "all" ? "Showing all courses." : "Showing " + category + " courses.";
}
function enroll(course){
  document.getElementById("message").textContent =
    "Demo: selected “" + course + "”. A real platform would sign you in and save enrolment.";
}
</script></body></html>
```

Conclusion

These three projects demonstrate common front-end development techniques: semantic HTML, responsive CSS layouts, navigation, visual effects, reusable cards, and JavaScript interactions. They can be expanded with real content, accessibility testing, image assets, form validation, and backend services as needed.

Suggested improvements

• Test keyboard navigation and colour contrast.
• Replace sample content and email addresses with approved details.
• Add real images with descriptive alternative text.
• Validate the layout on mobile, tablet, and desktop screens.
• For production use, add secure server-side handling for forms, admissions, user accounts, and enrolments.